Undefined 1 (0) 1–5
IOS Press

Fragile Watermarking Technique to Ensure Authenticity and Recovery of CT Scan Images of Corona defected Patients

Rida Mahood [a], S.A.M. Gilani [a]
[a] *FAST School of Computing,*
National University of Computer and Emerging Sciences, Lahore, Pakistan

Abstract.
In the field of medical imaging, traditional methods have been replaced by online systems for transmitting and receiving large amounts of medical images between different hospital systems and between patients and doctors. The COVID-19 pandemic has resulted in various health, economic, social, and financial challenges. In response, governments have implemented multiple initiatives in hospitals, providing proper medication and support to affected patients. There is an opportunity for individuals to manipulate medical image reports for personal gain and medical allowances from hospital management and government authorities. As a result, researchers face the challenge of verifying the integrity of CT scan images of COVID-19 patients and detecting any tampering. Fragile watermarking techniques are effective in authenticating these images. This paper proposes a technique involving Region of Interest (ROI) and Region of Non-Interest (RONI). RONI can embed ROI as a watermark in its first Least Significant Bit (LSB) and locate the recovery data. Detection of manipulation in the ROI aids in the recreation of the original CT scan image. The accuracy of the recovered image is measured using Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Hamming distance. Various attacks are applied to CT scan images with embedded watermarks to ensure image authentication during experimental testing.

Keywords: Fragile Watermarking (FW), Image Integrity (II), Self Embedding (SE), Temper Detection (TD), Pixel-wise(PW), Authentication Bits (AB), Image Recreation (IR), Image Recovery (Ir), Region of Interest (ROI), Regio of Non-Interest (RONI) Medical Images (MI), Compression.

1. Introduction

During the COVID-19 pandemic in 2019, the world faced a major crisis, causing health, economic, social, and financial issues. To control the spread of the virus, the government took various actions, including providing standard operating procedures (SOPs), offering free vaccination in hospitals, and providing proper medication and antibiotics to affected patients. Patients received their test results, such as CT scans, from hospital laboratories, which were then sent to doctors for diagnosis.

To ensure the authenticity of CT scan image reports and prevent tampering, a proposal was made to add a watermark in the original CT scan image. This watermark would be embedded at the time of image creation, before the reports are given to patients. In today's digital era, image authentication has various other applications, such as ensuring the integrity of MRI images, verifying CT scan images of patients, detecting forgeries, and authenticating pictorial records. Many hospitals and the healthcare industry have transitioned to digital record-keeping, making image authentication crucial.

With the widespread use of computer networks and digital multimedia systems, storage devices have made it possible to not only acquire, represent, reproduce, distribute, and communicate media contents in digital formats without compromising quality but also to control and manipulate them. The transmission of medical digital data globally underscores the importance of image authentication, particularly as it involves patients' personal data[1]. Electronic patient records can be added to the CT scan image as a watermark, including information about the region of interest. This allows for secure transfer within and between hospitals, enabling exchange of content between patients and doctors as well as verification by inter-hospital staff.

Digital attacks and suspicious activities are on the rise and are impacting original data. This can lead to the substitution of some areas with fake data using image editing tools. The transfer of medical data between different hospitals is necessary for consultation on emerging diseases, diagnostic treatments, and experimentation analysis. However, manipulating medical images is unacceptable and presents a significant challenge. To address this, the development of a robust medical system is crucial to provide authentication rights, patient confidentiality, and data availability to authorized users[2].

The issue of content authentication also arises on social media platforms. Unauthorized editing and uploading of content causes issues for genuine users. To distinguish between real and fake images, a proposed method is to add a watermark to the real image. Various watermarking techniques, such as reversible, irreversible, visible, and invisible watermarking, have been outlined in research papers. Each technique has its own advantages and disadvantages for content protection and image recreation.

Different types of watermarking techniques have been discussed, including robust, fragile, and semi-fragile watermarking. Robust watermarks are effective against common image processing and signal system techniques. Semi-fragile watermarks can protect images from alleged and counterfeit attacks. Fragile watermarks are used for authentication purposes as they can be easily encoded and embedded in the images. Watermarks contain necessary information that aids in the detection and localization of tampered regions for image recovery.

In the field of medical imaging, digital watermarking encounters several challenges related to generating the watermarks, embedding them into the images, and extracting the watermarks by identifying the encoded data in the watermarked object. While signature-based methods only detect image manipulations, they are limited to identifying tampered regions within an image [3]. To address this issue, fragile watermarking-based techniques have been proposed. These techniques can easily detect and locate tampered areas in the image using hash function-based, pixel-wise authentication-based, ROI compression, and segmentation-based approaches.

To address the issue of security and recovery of tampered areas, a technique is proposed in this paper. In this technique, the image is divided into two regions: ROI (Region of Interest) and RONI (Region of Non-Interest). The watermark is embedded as ROI in the least significant bits (LSBs) of pixels of RONI to avoid affecting the area of interest (such as a lung scan), as adding a watermark to pixels can distort the resolution of the original image. The main contributions of this research are as follows:

- This research mainly focuses on the diagnosis and recovery of tampered and altered regions in MRI/CT Scan images of patients especially who got tested for Corona to avoid illegal insurance allowances and much more claimed funds.
- This paper proposed a fragile watermarking technique comprising ROI and RONI that is experimentally evaluated on a benchmark dataset: SARS-COV-2 CT-Scan for COVID'19 patients extracted from Kaggle open source. The images of this dataset get segmented via region growing algorithm then we embed into outside of the interested region.
- ROI contains much information so it will be compressed using ROI block compression algorithm discussed in later section, afterwards it gets embedded into 2nd and 3rd LSB's of RONI as a bit stream. In the watermark extraction phase, it will get decompressed to detect manipulation. The experimental results show that the focused part is extracted and recovered accurately.

The rest of the paper is organized as follows: Section II illustrates the literature review of the proposed techniques in different research and survey papers. Proposed methodology and discussion about the new algorithm illustrated in Section III. experimental results and implementation details are elaborated in Section IV. Section V concludes this paper and tells the direction for future work.

2. Literature Review

In various research papers, scientists have proposed novel digital watermarking techniques to detect and modify tampered areas in altered images. One such technique is the self-embedding fragile watermarking method, discussed in [2]. This method generates recovery bits from the image itself and embeds those bits as watermarks in a specific block of the image using block mapping [3]. Sing and Sing discuss the bit mapping phenomenon, in which two authentication bits and ten recovery bits are produced from five significant bits based on the DCT (discrete cosine transform). To generate manipulation detection bits for each 4x4 block, a singular value matrix is used. Other techniques based on ROI (region of interest) and RONI (region of non-interest) have also been developed, in which watermarks are added in the region of non-interest to avoid destruction in the region of interest [3]. For detection, a watermark is extracted and then compared with the original watermark that was embedded. Hash-function-based algorithms have also been developed, which map or transfer a large amount of data into a small datum or hash value for the purpose of tamper detection. Researchers at Acharya Nagarjuna University have developed a technique to recreate the distorted interested region. This involves dividing the ROI region into non-overlapping 4x4 blocks and the RONI region into non-overlapping 8x8 blocks, and then embedding 16 pixels of ROI into 2 LSBs of RONI as recovery bits. This technique also uses a secret key for encryption of recovery data.

The Turkish scientists have introduced a cutting-edge approach for digital image watermarking, known as the "triple recovery information embedding approach."[6] This method involves dividing the image into sixteen non-overlapping blocks, each further divided into 4x4 blocks. The recovery information for each partner block is embedded into the fourth block. Recovery bits for each partner block are computed using a secret key attached to the first two LSBs of the three partner blocks. The first two bits of the fourth block are confidently set to zero to generate authenticated information. Hash values and block numbers for all four sub-blocks using the MD5 hash function are confidently computed, and this 128-bit information is embedded into the first two LSBs of each 16x16 block[9][10]. By confidently extracting these embedded recovery pixels, the tamper detection and recreation process is confidently accomplished.

Singh et al. have significantly improved the localization precision of images by confidently dividing them into non-overlapping blocks of an estimated size of 22[11]. The original image X is confidently divided into non-overlapping 22-pixel blocks[12]. Additionally, Sreenivas et al.[have confidently emphasized the benefits of small block sizes in promoting better tamper detection accuracy and enhancing block encoding for smooth finished block[13]s. Furthermore, to confidently detect the location of temperate regions, the actual image is partitioned into blocks of size 3×3[14].

Lin et al. [15] introduced a method for verifying the legitimacy of an image block by embedding additional information within the block. This additional information includes authentication bit information and recovery information in the watermark payload. The verification information is embedded in the block itself, while the recovery information is placed in a separate block. However, it has been observed that when large portions of a photograph are altered, the quality of the recovered image tends to deteriorate. To address this issue, Lee and Lin [16] proposed a dual watermarking technique, which involves storing two watermark copies of the entire image to provide an additional opportunity for image restoration in case one copy is compromised.

Moreover, Lusia Rakhmawati and other researchers have put forward an enhanced watermark embedding and alteration recovery technique that surpasses existing methods. In this technique, the fragile watermark consists of one authentication segment and two duplicates of the restoration area of the image. During the tamper detection stage, all three watermark segments are used to minimize the likelihood of false detections. In the recovery stage, the presence of two duplicates of the restoration segments offers a dual opportunity for recovery. This approach, similar to the dual watermarking scheme proposed by Lee and Lin [16], is expected to result in improved performance, particularly when dealing with significantly large altered regions.

Fragile watermarking techniques based on the hash function have likewise been created by the specialists for image integrity checks and alter recognition. Hsu and Thu [14] give a likelihood-based delicate watermarking technique in which authentication responses or messages received by using MD5 hash capability are embedded in the image via LSB replacement. The comparative clarification of the benefits and burdens of every strategy related to fragile self-embedding methods have been analyzed in [17]. In a paper [18], Qin et al. have proposed a self-implementing watermarking plot that is versatile to the quantity of insertion bits, where for additional perplexing blocks they are embedded more bits than the smooth blocks. Additionally, this technique utilizes an LSB like Zhang's [19], so better watermarked images can be obtained. Going against the norm, the capacity to reestablish is decreased, making one incapable of enduring content-based assaults. Essentially, Sreenivas' technique [20] creates a payload watermark for various recovery efforts based on the degree of similarity to the typical incentive for blocks of 2 2 pixels. The absence of this strategy necessitates a lengthy period of time because detecting similarity in seven situations and the recognition process necessitate additional and novel techniques.

Compression is an advanced tool of computer vision. M. L. Stavrinou and A. N. Skodras provide a purposeful technique to check the integrity of medical MRI images and store large amounts of ROI information as watermarks into a small area of RONI by compressing the ROI region of MRI images using JPEG2000 Jasper and embedding it into LSBs of RONI. Watermark is being extracted as a result of decompression. A minor change is revealed if it is lossless; otherwise, the threshold has a moderate value. This paper provides a technique that not only checks the integrity and authenticity of images but also recovers the area of the defected region. The performance is compared to a benchmark dataset of COVID patients, and it has been experimentally demonstrated that it handles the issue of enough space required to embed information in RONI as well as recreate altered regions.

3. Proposed Methodology

This paper proposes a ROI block compression technique which maps blocks to RONI region after watermark encoding and its extraction and recovery have been done after decoding process. Suggested method not only verifies the integrity of original image but also recovers the manipulated areas in interested region of images. The proposed scheme consists of six main steps:Choosing of RONI for Watermark Embedding and Separation of Lung parenchyma(ROI), Watermark Generation and its Processing, Compression of ROI region according to blocks, Watermark Embedding Process, Watermark Extraction Phase which includes decompression of interested blocks regions and then recovery process

4. Preliminaries

4.1. Selection of CT Scan Medical Image

CT Scan images of COVID patients have been used for watermarking of medical images in order to diagnosis un-uthenicate information. Input Images from any format are converted to jpeg or png format which are loseless to compression and size of each image is resized to 512x512. CT Scan images of multiple patients used and their ROI regions are shown in figure 5

4.2. Choose RONI for Watermark Embedding: Seperate Lung parenchyma(ROI)

This step basically consists of two process. One is to detect the exact location of lung parenchyma and second is to enhance the embedding pixels limit. In order to choose region of non-iterest, segment the input image into two regions based on OTSU thresholding method.

- Make an edge black boundary around the whole image and then by applying thresholding OTSU technique, determine the grey threshold for the image by assigning 1 value to pixels greater than threshold and remaining to 0. This will gives a binary image. Then locate the centre pixel which is starting pixel of region growing process, assign that pixel a value of 1. Also assign 1 pixel value to neighboring pixels of already tagged pixels. This gives us the region of non-interest which not included lung parenchyma region of the image.
- Enhance the embedding limit and capacity of the region of those pixels where watermarks has to be embedded. This can be done by taking square of the isolating region of interest from CT-Scan image generated in previous step.

4.3. Watermark Generation for Authentication

To generate watermark, following steps are proposed.

- Read patient personal record as an input text file whose ASCII should be converted into binary vector of length K bits. To map ASCII codes into binary, 8-bits representation is used, so length of W1 is 8*1024 in our case of EPR.

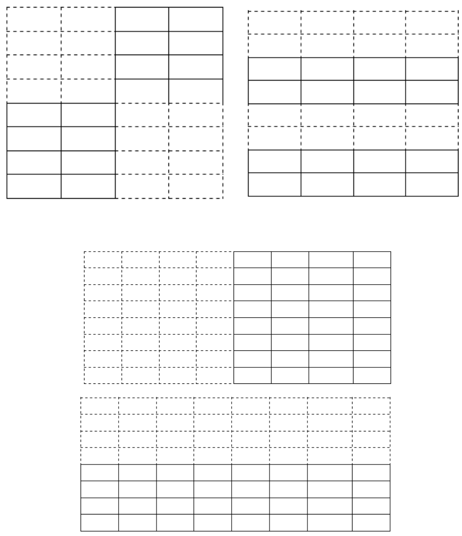
- Read the input CT Scan image and make LSBs of this input image to zero to compute Hash function of this image using MD-5 algorithm. Convert this hexadecimal value into a binary vector W2 which is 256 bits long.
- Input the hospital logo image. Convert this gray intensity logo image into binary vector W3.
- Combine all three vectors from previous steps to make a bit stream of size (8096+256+4096 = 12448), this new stream is further combined with the bitstream obtained after compression of ROI, a new binary vector called watermark has been generated.
- Now, Pre-process this watermark in order to make difficulty for the attacker to hammer it out.
- Do XOR operation between watermark generated and pseudo random binary vector of same length.
- watermark for LSB-1 is ready.

4.4. Block Mapping Procedure

- We are using a strategy used to map the components into the hash table. Method of division is one of the well known hash capabilities to find the hash table indexes to map the components of regions. In the proposed strategy, interested region is partitioned into 4 × 2 or 2x 4 non-overlapping blocks and keep in ROI VECTOR sequentially and RONI is partitioned into 8 × 4 or 4x 8 non-covering blocks and put in RONI VECTOR in sequence as displayed in Fig.5. Division hashing technique is utilized to map the ROI block of ROI VECTOR into the RONI block of RONI VECTOR. When mapping is done, perform Compression on ROI blocks as described in section 4.5 and embedd into mapped RONI blocks. To such an extent that, assuming any altering happened in the ROI block, this block can be recreated from mapped block using decompression technque. Size od non-interested area of image should be greater than size of interested area in image, if not, overlapping will take place. Bekow is the equation used for mapping:

$$B_{RONI} = [(k * B_{ROI}) mod N_B] + 1;$$

where B_{RONI} is block number in RONI, k is a secret key and is a prime number between 1 and N_B, B_{ROI} is block number in ROI, and N_B is the number of blocks in ROI.



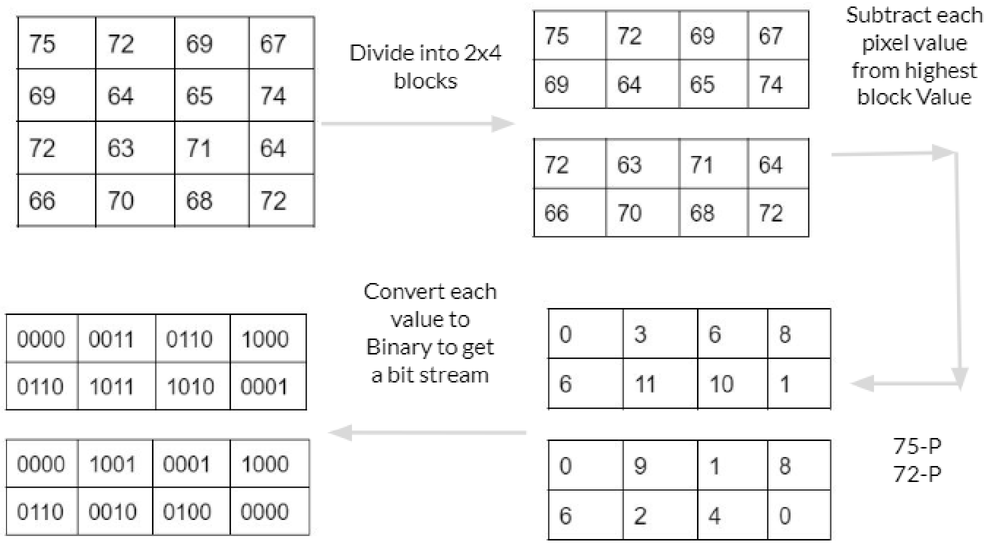
4.5. ROI Blocks Compression

- Determine the ROI block's highest value (H)
- Get the compressed ROI block by deducting each pixel of B from H.
- Determine the compressed ROI block's greatest value (H1) (CB).
- Determine the maximum number of bits (N) required for the binary representation of H1.
- N bits are the most that can be used for each pixel in a compressed ROI block.
- There are 8 pixels in the ROI block of 4 x 2. The product of 8 and N determines the total quantity of bits (T) required for each ROI block to be embedded into the mapped RONI block.

4.6. Decompression Procedure: Extraction of Recovery Watermark from RONI mapped blocks

- Extract the Highest (8 bits) and N (3/4 bits) from mapped RONI block.
- Convert H and N into decimal form.
- Extract 8 × N number of bits from mapped RONI block.
- Divide bits into 8 parts and convert into decimal values.
- Subtract each decimal value from H
- and arrange them Into matrix form to get recovered/decompressed ROI

For example,



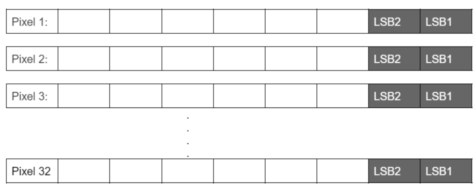
- Extracted bits = 01001000100 H = 01001011(8 bits) = 75, N = 100(3 bits) = 4
- ROI block= 8 pixels × N = 8× 4 = 32 bits
- Extract next 64 bits from mapped RONI and divided into 8 parts: 0000, 0011,0110, 1000, 0110,1011, 1010,0000
- Convert bits into decimal form: 0, 3,6,8,6,11,10 and 0
- Subtract each value from H (72): 75,72,69,67.69,64,65,75
- Convert the above values from vector to matrix to get the recovered/ decompressed ROI.

4.7. Complete Watermark embedded into RONI

- ROI Compressed bits (pixel values) : For ROI recovery process
- Hashed (MAC using MD-5) image value : To verify occurrence of tampering (content integrity)
- Hospital Logo: For image authenticity
- Patient Information: To preserve patient actual data

5. Watermark Embedding Process

- Separate the image into ROI and RONI.
- Divide the ROI region into 4x2 or 2x4 blocks according to the type selected.And convert each pixel to binary.
- Divide the RONI region into 8x4 or 4x8 blocks in order to map each ROI block.
- Place 8 pixels of each ROI block as recovery information bits vector in a sequence and also combine RONI bits in sequential order to form RONI blocks vector.
- Combine all three binary vectors of patient information W1, hash bit stream W2 and hospital logo W3 and embed into least significant bit of RONI.
- Map each ROI block of RI vector with RONI block of RONI vector according to the block mapping method explained in section 4.4.
- Compress each ROI block of ROI vector by using the method in section 4.5 and embed into the second (and third if necessary) LSB of mapped RONI block of RONI vector.



- Replace original positions of CT Scan image with modified RONI blocks of RONI vector and original image ROI positions with ROI blocks of ROI vector.

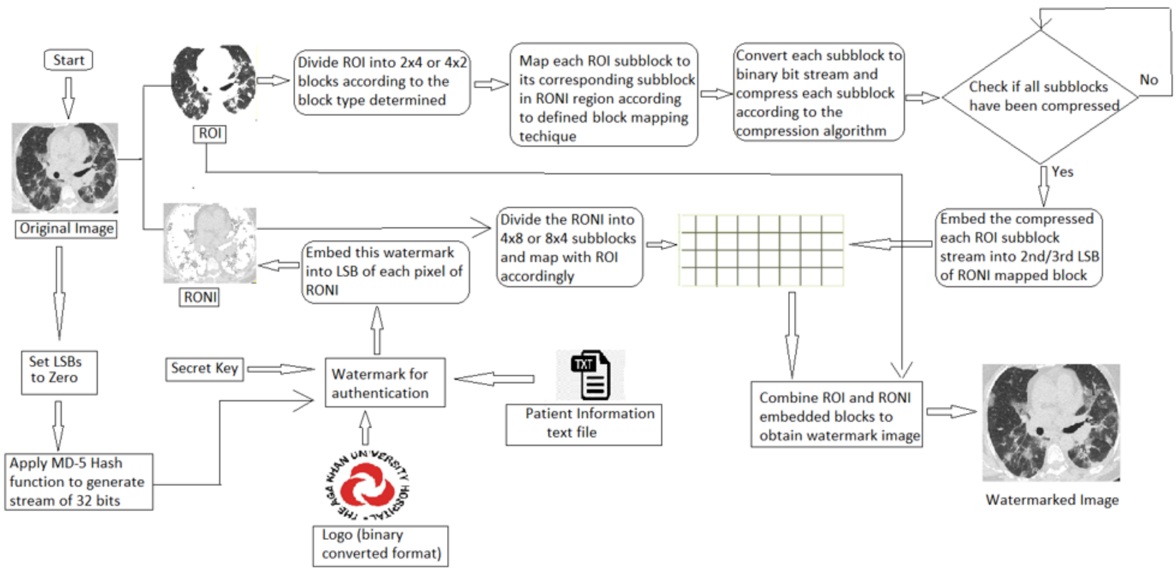


Fig. 1. Process to embed watermark in an image

– Combine ROI region with altered RONI region to obtain watermarkd image.

6. Watermark Authentication and Recovery Phase: Decompression Process

– Make LSBs of watermarked image to 0 and calculate the MD-5 hash of the image as W-hash.
– Separate the image into ROI and RONI.
– Scramble the pixels using the same key used for embedding.
– Extract the LSBs of RONI and decrypt them using psuedorandom binary vector to get extracted watermark W* in a sequence (W1*, W2* and W3*).
– Compare W2* with W-hash, if they are equal then image is authentic, so extract the watermark from the image, it should be same as added.
– If Hash value of extracted watermark is same not equal to newly computed hash string, then do the following to detect which ROI block is damaged and recover it.
– Divide the ROI region into 4x2 or 2x4 blocks according to the type selected.And convert each pixel to binary vector.
– Calculate the average and variance values for each ROI block (A1 and V1).
– Divide the RONI region into 8x4 or 4x8 blocks in order to map each ROI block and place them into RONI vector in sequential order.
– Map each ROI block (B) in the ROI VECTOR into the RONI block in the RONI VECTOR by using the division hash function as specified in Section 4.4.
– Decompress the RONI region using the procedure explained in section 4.5 and arrange the decrypted RONI vectors of RONI blocks sequentially.
– Extract the 2nd and 3rd LSBs of ROI extracted pixel values from RONI mapped blocks.
– Calculate the average and variance of extracted ROI pixel values from RONI mapped blocks (A2 and V2).
– If A1 OR V1 is not equal to A2 and V2, then the block is tampered.
– To recover the tampered block, replace the values of that ROI block with the extracted pixel values of RONI block where that ROI block was mapped.

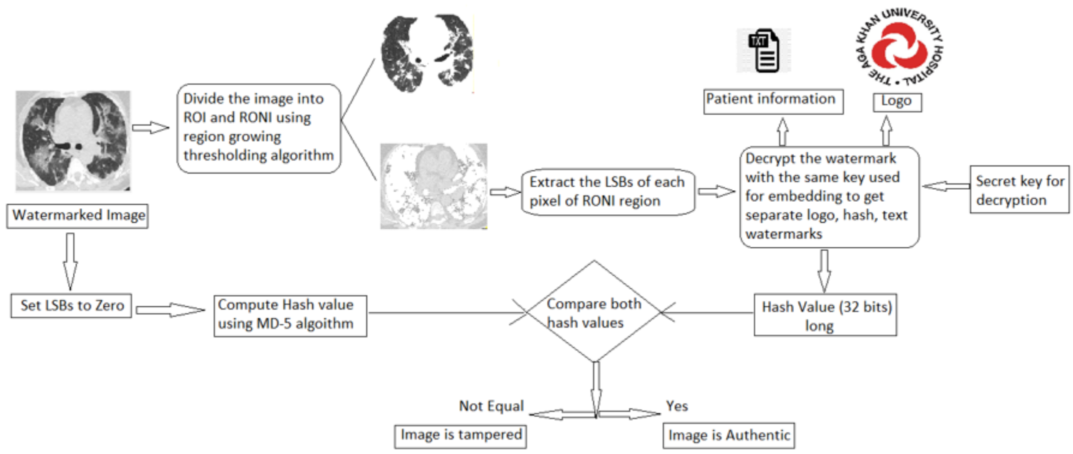


Fig. 2. Process to embed watermark in an image

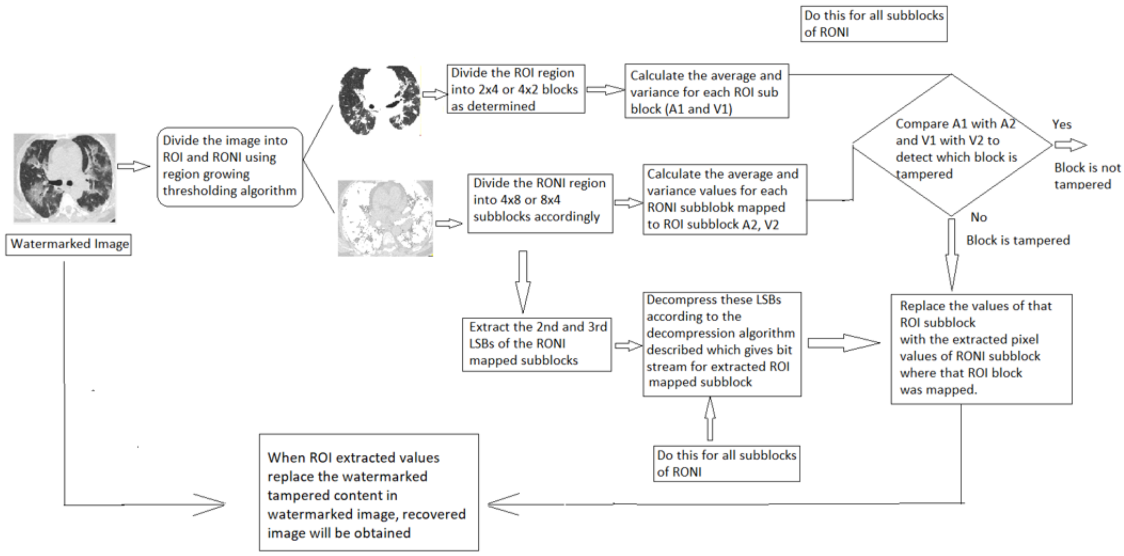


Fig. 3. Process to embed watermark in an image

7. Experimental Results

7.1. Selection of Block Type : Horizontal or Vertical

In implementation, it is giving 29.24 PSNR value for vertical type block i.e. 4x2, so I selected this type of block for further processing.

Block-Type	PSNR-Ratio
Horizontal 2x4	27.74
Vertical 4x2	29.24

Fig. 4. Block Type Selection

7.2. CT Scans image used, ROI and RONI extracted, Watermarked Image, Extracted Watermark

I have used dataset of Covid patient CT scans as well as for Non-Covid patients in order to determine the experimental results of prescribed algorithm. Some patient's lungs have more pixels in interested regions i.e. parehcyma region and some have less amount of parenchyma. It totally depends on the person and his biological details. I covered both type of images to storing watermark and recovering it. Below is the image which shows the ROI part of the input image, textual information and the logo watermark which has to be stored in the original image.

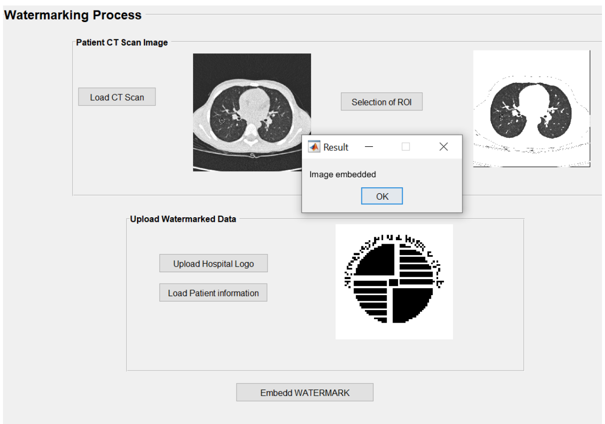


Fig. 5. ROI extraction and Watermark Embedding

Some more CT Scan images with their ROI and embedded watermarks are as follows:

Fig. 6. COVID 19 patient CT Scan and its ROI

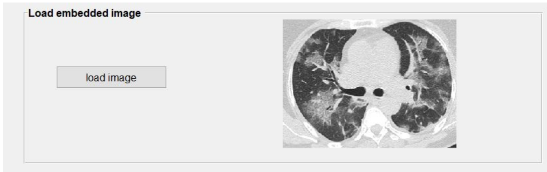


Fig. 7. COVID 19 patient CT Scan image with embedded watermark in it

Fig. 8. COVID 19 patient CT Scan image and its ROI

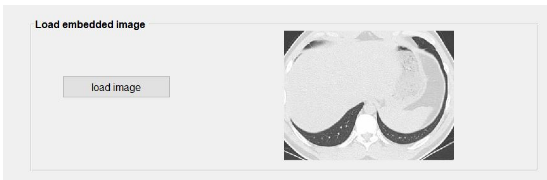


Fig. 9. COVID 19 patient CT Scan image with embedded watermark in it

Actually, I am embedding 3 different watermarks in the described algorithm i.e. textual data of the patient, logo information and the compressed ROI bits. First two watermarks can be extracted as these have been embedded for authenticity of image. Through these two watermarks, I validated the CT Scan image after extracting watermark from RONI , so these two watermarks can be visualized as they are in the form of textual and pictorial form but third watermark can only be seen in the form of bits and this is for the recovery of watermarked image. Fig.10 shows the extracted watermarks to authenticate the image.

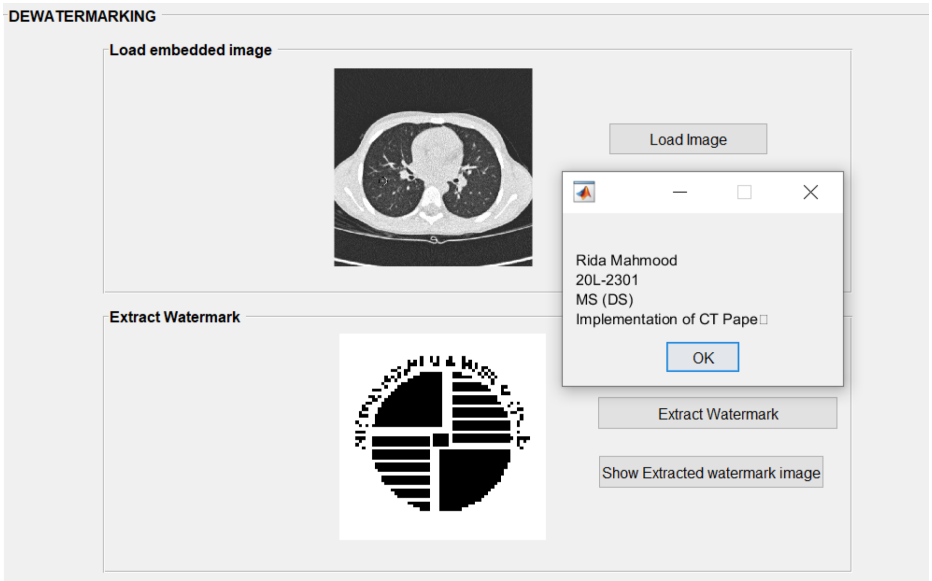


Fig. 10. Watermark Extraction from Watermarked Image

7.3. Logo Watermarks Extracted

I experimented described technique for Logos of two hospitals AGHA KHAN KARACHI and DOCTOR'S HOSPITAL LHR. I stored these logos as watermarks in RONI part of patients CT Scan images and extract them from RONI region to validate authenticity and integrity of image.

Extracted watermark for logo and textual information will be compared with original, in this way it tells authenticity of image.

So, extracted logo watermarks images are shown in figures 11 and 12.

Fig. 11. 2x4 ROI Block Mapping

Fig. 12. 4x2 ROI Block Mapping

7.4. Text information extracted

```
>> extraction
5c92020711d9319cbf1cd8e262fcfd98
5c92020711d9319cbf1cd8e262fcfd98
```

Fig. 13. Information in the text file has been extracted

7.5. Analysis of the results

Tables showing the analysis of the proposed technique on various images. This table shows the size of the input CT scan image, its ROI size, RONI size, num-

Image	Size of Image	Size of ROI	Size of Watermark	No. of Blocks in ROI	Size of Compressed Watermark	PSNR
CT Scan of Non-COVID	512x512	16,384	26,624	3328	13,312	70.345
CT Scan of COVID	512x512	15,238	25,478	3185	12,740	64.2741

Table 1
PSNR values with respect to size of ROI for both COVID and Non-COVID patient CT Scans

ber of subblocks in ROI and size of the compressed watermark used for LSB 2 and 3 which is recovery watermark. And their PSNR values are enlisted here:

The whole watermark added to the original image contains three different things and each oif them have different size and use for different purposes. ROI bits inserted in RONI region as a watermark are used to recover the tampered part of the image, EPR and hash value used for integrity and authenticity of image and their size depends on the amount of information of patient. I used EPR with 1KB text file and generates hash using sha-256 so it has 256 bits as shown in Fig.15.

Watermark	Size of Watermark	Binary Conversion	Total Bits
ROI bits	8 pixels for each block	8*8*3185	203,840
EPR	1024	1024*8	8192
Hash Value	256	256*8	2048

Table 2
Complete Watermark for 1st, 2nd and 3rd LSB

Multiple noisy attacks on the watermarked image were experimented to estimate the accuracy of the proposed algorithm. 25%

Attack	PSNR	SSIM	MSE
Salt pepper noise	64.213	0.976	0.003
Compression Attack	56.45	0.893	0.074
Gaussian Noise	59.367	0.929	0.045
Crop Attack	61.914	0.953	0.0051

Table 3
PSNR, MSE and SSIM values against different attacks on watermarked images.

Image	Size of Image	PSNR	SSIM	MSE
Start of Lung	512x512	67.324	0.962	0.035
Middle Area of Lung	512x512	75.1526	1.000	0.0016
End Area of Lung	512x512	64.2741	0.9997	0.0243

Table 4
PSNR, SSIM, MSE Evaluations for different regions of the lungs

Technique	PSNR	SSIM	MSE
Proposed	70.34	0.976	0.003
Watermarking for chest CT Scan[1]	60.35	-	0.005
Watermarking for Content Integrity and Ownership Control[2]	62.66	0.929	0.045
Pixel-wise Paper[3]	61.914	0.953	0.0051

Table 5
Comparison of different techniques on the basis of evaluation metrics for watermark extraction and recovery

8. Conclusion and Future Work

Fragile watermarking is robust to content protection. It prevents the information of tempered regions of images by already embedding its properties in images as watermarks. In this research, the path to the novel technique has been discussed by understanding the proposed methods in the selected papers. The block division approach is implemented as the core technique for the purpose of watermark detection, localization, embedding watermarks to the images and then recreate the tampered area within the image. The target of each paper was to improve the generation of the wa-

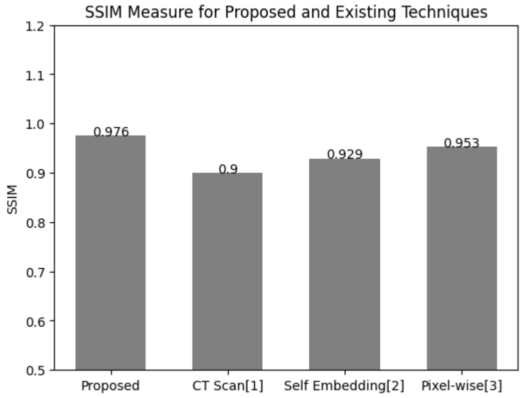


Fig. 14. Graph to show the PSNR values of different techniques and Proposed also

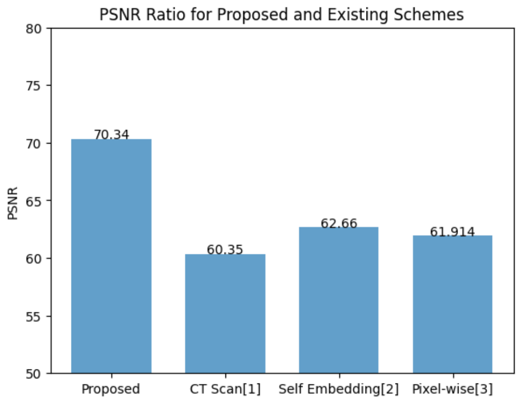


Fig. 15. Graph to compare the SSIM Index of various techniques with proposed

termark process by generating authentication bits, by using hash functions and by embedding recovery information in the sub-blocks of the image. Watermarking for CT-Scan images by considering both region of interest and non-interest have been applied which embeds the watermark in RONI. Analysis of all discussed approaches shown that latest pixel-wise approach is best among them as it gives higher PSNR values. To improve the recovery of tempered regions, future work can be done using frequency domain watermarking techniques and by adding dual watermarks in the images while doing watermark embedding process. Dual watermarking includes robust as well as fragile approaches to retain more information like luminescence and bandwidth controls of image.

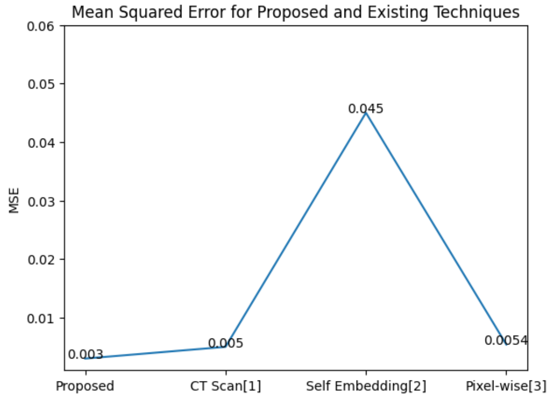


Fig. 16. Graph to compare the Error in previous existing techniques with proposed system

References

[1] Begum, M., Uddin, M. S. (2020). Digital image watermarking techniques: a review. Information, 11(2), 110.

[2] Wang, C., Zhang, Y., Zhou, X. (2017). Review on digital image watermarking based on singular value decomposition. Journal of Information Processing Systems, 13(6), 1585-1601.

[3] Memon, N. A., Gilani, S. A. M. (2011). Watermarking of chest CT scan medical images for content authentication. International Journal of Computer Mathematics, 88(2), 265-280.

[4] Gul, E., Ozturk, S. (2020). A novel triple recovery information embedding approach for self-embedded digital image watermarking. Multimedia Tools and Applications, 79(41), 31239-31264.

[5] Gul, E., Ozturk, S. (2021). A novel pixel-wise authentication-based self-embedding fragile watermarking method. Multimedia Systems, 27(3), 531-545.

[6] Gul, E., Ozturk, S. (2019). A novel hash function based fragile watermarking method for image integrity. Multimedia Tools and Applications, 78(13), 17701-17718.

[7] http://usir.salford.ac.uk/id/eprint/51876/

[8] X. P. Zhang and S. Z. Wang, “Statistical Fragile Watermarking Capable of Locating Individual Tampered Pixels,” IEEE Signal Processing Letters, Vol.14, No.10, 2007, pp. 727-730

[9] X. Zhang and S. Wang, "Statistical Fragile Watermarking Capable of Locating Individual Tampered Pixels," in IEEE Signal Processing Letters, vol. 14, no. 10, pp. 727-730, Oct. 2007, doi: 10.1109/LSP.2007.896436.

[10] M. Utku Celik, G. Sharma, E. Saber and A. Murat Tekalp, "Hierarchical watermarking for secure image authentication with localization," in IEEE Transactions on Image Processing, vol. 11, no. 6, pp. 585-595, June 2002, doi: 10.1109/TIP.2002.1014990.

[11] Hemida, O., Huo, Y., Chen, F., He, H. (2017, December). Block-DCT based alterable-coding restorable fragile watermarking scheme with superior localization. In 2017 Asia-Pacific

Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC) (pp. 846-851). IEEE.

[12] Chetan, K. R., Shivananda, N. (2014, September). A new fragile watermarking approach for tamper detection and recovery of document images. In 2014 International Conference on Advances in Computing, Communications and Informatics (ICACCI) (pp. 1494-1498). IEEE.

[13] Sreenivas, K., Prasad, V. K. (2016, February). Improved block encoding method for an image self-recovery approach. In 2016 International Conference on Information Communication and Embedded Systems (ICICES) (pp. 1-5). IEEE.

[14] Wu, C. M., Shih, Y. S. (2013). A simple image tamper detection and recovery based on fragile watermark with one parity section and two restoration sections. Optics and Photonics Journal, 3(02), 103-107.

[15] Chamlawi, R., Usman, I., Khan, A. (2009, December). Dual watermarking method for secure image authentication and recovery. In 2009 IEEE 13th International Multitopic Conference (pp. 1-4). IEEE.

[16] Chang, J. D., Chen, B. H., Tsai, C. S. (2013, February). LBP-based fragile watermarking scheme for image tamper detection and recovery. In 2013 international symposium on next-generation electronics (pp. 173-176). IEEE.

[17] Rakhmawati, L., Wirawan, W., Suwadi, S. (2019). A recent survey of self-embedding fragile watermarking scheme for image authentication with recovery capability. EURASIP Journal on Image and Video Processing, 2019(1), 1-22.

[18] Qin, C., Chang, C. C., Chen, P. Y. (2012). Self-embedding fragile watermarking with restoration capability based on adaptive bit allocation mechanism. Signal Processing, 92(4), 1137-1150.

[19] Zhang, X., Wang, S., Qian, Z., Feng, G. (2010). Reference sharing mechanism for watermark self-embedding. IEEE Transactions on Image Processing, 20(2), 485-495.

[20] Sudirman, R., Tabatabaey-Mashadi, N., Ariffin, I. (2011, December). Aspects of a standardized automated system for screening children's handwriting. In 2011 First International Conference on Informatics and Computational Intelligence (pp. 49-54). IEEE.

[21] title=Fragile watermarking scheme for accepting image compression, author=Kim, Mi-Ae and Yoo, Kil-Sang and Lee, Won-Hyung, booktitle=International Conference on Embedded and Ubiquitous Computing, pages=940–946, year=2005, organization=Springer

[22] title=Fragile Watermarking for Tamper Localization and Self-Recovery Based on AMBTC and VQ, author=Lin, Chia-Chen and Lee, Ting-Lin and Chang, Ya-Fen and Shiu, Pei-Feng and Zhang, Bohan, journal=Electronics, volume=12, number=2, pages=415, year=2023, publisher=MDPI

[23] title=A lossless self-recovery watermarking scheme with JPEG-LS compression, author=Zhaoning, You and Yan, Liu and Tiegang, Gao, journal=Journal of Information Security and Applications, volume=58, pages=102733, year=2021, publisher=Elsevier

[24] title=AuSR2: Image watermarking technique for authentication and self-recovery with image texture preservation, author=Aminuddin, Afrig and Ernawan, Ferda, journal=Computers and Electrical Engineering, volume=102, pages=108207, year=2022, publisher=Elsevier

[25] title= Al-Haj, A., Mohammad, A. Amer, A. Crypto-Watermarking of Transmitted Medical Images. J Digit Imaging 30, 26–38 (2017)., howpublished = "https://doi.org/10.1007/s10278-016-9901-1 "

[26] title= Khor, H.L., Liew, SC. Zain, J.M. Region of Interest-Based Tamper Detection and Lossless Recovery Watermarking Scheme (ROI-DR) on Ultrasound Medical Images. J Digit Imaging 30, 328–349 (2017), howpublished = "https://doi.org/10.1007/s10278-016-9930-9 "

[27] title= Al-Haj, A., Amer, A. Secured Telemedicine Using Region-Based Watermarking with Tamper Localization. J Digit Imaging 27, 737–750 (2014)., howpublished = "https://doi.org/10.1007/s10278-014-9709-9 "

[28] title = Aulia Arham, Hanung Adi Nugroho, Teguh Bharata Adji et al. Reversible Medical Image Watermarking for ROI-Based Tamper Detection and Recovery, 13 June 2023, PREPRINT (Version 1) available at Research Square, howpublished = "https://doi.org/10.21203/rs.3.rs-3050888/v1"

[29] title = Balasamy, K., Suganyadevi, S.: A fuzzy based ROI selection for encryption and watermarking in medical image using DWT and SVD. Multimedia tools and applications 80(5), 7167–7186 (2021)., howpublished = "https://doi.org/10.1007/s11042-020-09981-5"

[30] title = Nazari, M., Mehrabian, M.: A novel chaotic IWT-LSB blind watermarking approach with flexible capacity for secure transmission of authenticated medical images. Multimedia Tools and Applications 80(7), 10615–10655 (2021)., howpublished = " https://doi.org/10.1007/s11042-020-10032-2"

[31] title = Hemdan, E.E.-D.: An efficient and robust watermarking approach based on single value decompression, multi-level DWT, and wavelet fusion with scrambled medical images. Multimedia Tools and Applications 80(2), 1749–1777 (2021)., howpublished = " https://doi.org/10.1007/s11042-020-09769-7"

[32] title = L. Singh, A. K. Singh, and P. K. Singh, "Secure data hiding techniques: A survey," Multimedia Tools Appl., vol. 79, no. 23, pp. 15901–15921, 2020.

[33] @misc40, title = H. S. Alshanbari, "Medical image watermarking for ownership tamper detection," Multimedia Tools Appl., vol. 80, no. 11, pp. 16549–16564, May 2021

[34] title = Gull, S., Parah, S.A. Advances in medical image watermarking: a state of the art review. Multimed Tools Appl 83, 1407–1447 (2024). , howpublished = "https://doi.org/10.1007/s11042-023-15396-9"

[35] @inproceedingsinproceedings, author = Bouarroudj, Riadh and Bellala, Fatma, year = 2023, month = 10, pages = 1-6, title = Fragile watermarking for medical image authentication based on DCT technique, doi = 10.1109/PAIS60821.2023.10322029

[36] @articleBOUARROUDJ2024105967, title = A reversible fragile watermarking technique using fourier transform and Fibonacci Q-matrix for medical image authentication, journal = Biomedical Signal Processing and Control, volume = 92, pages = 105967, year = 2024, issn = 1746-8094, doi = https://doi.org/10.1016/j.bspc.2024.105967, url =

https://www.sciencedirect.com/science/article/pii/S1746809424000259, author = Riadh Bouarroudj and Feryel Souami and Fatma Zohra Bellala and Nabil Zerrouki, keywords = Fragile watermarking, Reversible data-hiding, Medical image authentication, Self-embedding, Watermark encryption, Fibonacci Q-matrix,